

# 980 SOFTWARE DOCUMENTATION

P/N 944102-9901

[illegible]

**TI-00023.C**

## I. Patch Procedure for Standard Operating Systems

This document provides the information necessary to update, by means of a series of patches, the DX980 Operating System previously shipped. These patches are to be incorporated into the operating system by means of the dump and modify disc commands. Users not familiar with these commands should reference Section II.

For a standard system as received from Texas Instruments the information provided in Table 1 and the printout of Figure 1 is all that is needed to implement the patches. Table 1 provides the following information about each of the patches: a description of the nature of the fix, the absolute sector address and the word offset within the sector for the patch, the present contents of the word and the patch to be applied to the word. Figure 1 provides a complete system console printout of the patch implementation process performed at Texas Instruments. The user may reproduce this printout at his installation to implement the patch update process on his system. He need only re-IPL following this procedure in order to insure that the patched system is being utilized in memory.

For a nonstandard, or custom, system the procedure for patching the system consists of first calculating the absolute sector address for the patch and then following the standard procedure for applying the patch to that sector. The information necessary for calculating the address of the sector to be modified is provided the nonstandard user in Section III.

Nonstandard operating systems may result in several different ways.

1). The DX980 Source Kit (P/N 944140-0010) customer may have generated a custom DX980 Operating System.



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2). The customer may have copied (by means of the General Copy Utility DXCOPY) the load module affected by the patch.

3.) The customer may have modified (by means of the Load Module Update Utility LMUPDT) an individual MIP affected by the patch. This would alter the disc address at which the MIP starts and in addition could alter the word offset at which the patch should be applied. This latter case must be handled on an individual basis by the customer and cannot be treated in Section III.

If the DX980 system received from Texas Instruments has been modified in any of the three ways described above, please reference Section III. before applying the patches.



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## II. Use of Dump Disc and Modify Disc Commands

Disc patches may be applied with the aid of the Dump Disc and Modify Disc operator communications commands. First, to display the disc sector to be modified a Dump Disc command is entered on the system console. This command is of the form:

```
//DD DISNO SSEC ESEC
```

DISNO is an integer that specifies the disc drive number. Disc number 1 is always the system disc. For others discs, this number may be determined by the List Devices IPL command. The disc number is indicated in the ID column of this display.

SSEC is the starting sector number to be displayed. Sector numbers may be decimal ( $dddd \leq 32767$ ) or hexadecimal ( $>hhhh \leq 7FFF_{16}$ ), where hexadecimal sector numbers are preceded by  $>$ .

ESEC is the ending sector number being requested.

The display sector should be verified to match the listing before the patch is applied.

The patch is applied with the Modify Disc command.

```
//MD DISNO SSEC WRDNO DATA
```

WRDNO is the word off-set (either decimal or hexadecimal) into the sector to which the patch is to applied. Word offsets range from 0 to  $31_{10}$  or 0 to  $1F_{16}$ .

DATA represents one or more decimal or hexadecimal numbers separated by blanks. These will be applied to the disc sector consecutively starting at the word offset WRDNO.

After the patch is made the Dump Disc command should again be used to verify the changes.



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### III. Patch Procedure for Nonstandard Operating Systems

To install a patch to a nonstandard operating system, the customer uses the standard procedure except the correct disc address to be modified must be calculated. This address can be calculated from the information provided in Table II for each of the patches. The formula for calculating the absolute disc address is provided at the bottom of Table II. The procedure is as follows:

- 1) Run CATFIL to find the starting track numbers of the load modules listed in Table II.
- 2) Convert the starting track address for the particular load module listed for the patch to a starting sector address by multiplying by  $58_{16}$ . Enter this sector address in the blank column in the Table labeled "Load Mod Start Sector LS."
- 3) The MIP number in Table II indicates the MIP, or overlay, to be modified. If MIP=0, the patch is to the root segment itself and steps 4 through 7 can be skipped.
- 4) Add the relative sector number of the MIP directory words for the particular MIP (Shown in Table II under column labeled "Directory Words, Sector Offset DS") to the starting sector address of the load module.
- 5) Display the sector number calculated in Step 4.
- 6) The directory words for the MIP to be patched start at the word number indicated under column labeled "Directory Words, Word Offset DW." The first word of the directory words gives the sector number for the start of the MIP relative to the beginning of the load module.



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7) Add this relative starting sector address to the starting sector address of the load module. The result is the starting address of the MIP.

8) Add to the starting sector address for the MIP (this is equal to the address of the load module if the MIP number is 0) the sector offset within the MIP that is to be patched. Enter this absolute sector address to be patched in the blank column in the table labeled "Calc Abs Sector AS."

9) Display the sector number calculated in step 8. The contents of the displayed sector should match the display of the equivalent sector on the released DX980 system (see Figure 1).

10) Modify the same words in the sector calculated in step 8 as were modified in the released system. The word(s) to modify start with the word indicated in the column labeled "Word Offset into Sector". The present contents of the word(s) to modify and the patch to be applied are provided in the column labeled "Present" and "Patch", respectively.

11) Repeat steps 2 through 10 for each patch to be applied.

12) Re-IPL the patched operating system to ensure that the modified code is used.

As an example of applying a patch to a nonstandard operating system, the procedure is used to apply one of the patches for the Revision \*\* DX980 Operating System. The module being modified will be one from REV \*\* DX980 rather than a nonstandard DX980. The excerpt from Table II applicable to this example is reproduced here.



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| Patch # | Load Module | Load Mod Start Sector LS | MIP # | Directory Words  |                | Sector Offset Into MS | Calc Abs Sector AS | Word Offset into Sector | Present                 | Patch                   |
|---------|-------------|--------------------------|-------|------------------|----------------|-----------------------|--------------------|-------------------------|-------------------------|-------------------------|
|         |             |                          |       | Sector Offset DS | Word Offset DW |                       |                    |                         |                         |                         |
| 1       | SYSLD       |                          | >C    | 1                | 2              | >D                    | -                  | >D<br>>E<br>>F          | >8102<br>>0700<br>>8405 | >2102<br>>8102<br>>9405 |

Step 1. The CATFIL listing shows SYSLD to start on track  
Number  $13B_{16}$ .

CATALOG OF FILES FOR USER ID SYSTEM 1 / 1 / 74, 2 : 35: 40  
DISC VOLUME # 1

| FILE NAME | DISC ADDR | PASS WORD | INTEGRITY CODE | CURRENT FILE SIZE | MAX FILE SIZE | FILE TYPE | LRECL/ KEYLEN | PRECL WORDS |
|-----------|-----------|-----------|----------------|-------------------|---------------|-----------|---------------|-------------|
| SMR       | 0114      |           | 7,4,4,7        | 0005              | 0005          | RELREC    | 0040          | 0020        |
| SMRWRK    | 0021      |           | 7,7,7,0        | 0002              | 000A          | LINKSEQ   |               | 0080        |
| SDQ       | 003E      |           | 4,4,4,4        | 0001              | 0001          | RELREC    | 0040          | 0020        |
| SYSLD     | 013B      |           | 4,4,4,4        | 0020              | 0020          | RELREC    | 0040          | 0020        |
| USRFTN    | 00EA      |           | 7,4,4,0        | 000E              | 0014          | INDEXED   | 0008          | 0080        |
| USRPLX    | 0009      |           | 7,4,4,0        | 0008              | 0014          | INDEXED   | 0008          | 0080        |

Step 2. The starting sector address of SYSLD is calculated as:

$13B_{16}$  (Starting track no. of SYSLD).

$\times 58_{16}$  (Sectors/Track).

$6C48_{16}$  (Starting sector address of SYSLD).

Step 3. The MIP to modify is number  $C_{16}$ . Continue to step 4.

Step 4. The starting sector address of the MIP Directory is calculated as:

$6C48_{16}$  (Starting sector address of SYSLD).

+ 1 (Sector offset for MIP  $C_{16}$  Directory Words).

$6C49_{16}$  (Sector address for MIP Directory Words).



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Step 5.

//DD 1 >6C49  
6C49

|      |      |      |      |      |      |      |      |                   |
|------|------|------|------|------|------|------|------|-------------------|
| 005F | 0008 | 0211 | 0259 | 0028 | 0226 | 040E | 004F | .....Y.C.&.N.D    |
| 0250 | 012A | 0015 | 025B | 000B | 0010 | 0263 | 004B | .P.♦....C.C.....K |
| 0007 | 0267 | 0145 | 0017 | 0273 | 045B | 0048 | 0299 | .....E.....L.H..  |
| 0087 | 000B | 029F | 008C | 000B | 02A5 | 0181 | 001B | .....%.....       |

Step 6. The directory words for MIP  $C_{16}$  are indicated by the box in step 5.

Step 7. The starting sector address of the MIP is calculated as:

$6C48_{16}$  (Starting sector address of SYSLD).

$+211_{16}$  (Starting sector address of MIP relative to SYSLD).

$6E59_{16}$  (Starting sector address of MIP  $C_{16}$ ).

Step 8. The sector address to patch on the disc is calculated as:

$6E59_{16}$  (Starting sector address of MIP  $C_{16}$ ).

$+ D_{16}$  (Offset in MIP from Table B-1).

$6E66_{16}$  (Absolute disc sector address to be patched).

Step 9. The sector as it is currently on the disc is:

//DD 1 >6E66  
6E66

|      |      |      |      |      |      |      |      |                  |
|------|------|------|------|------|------|------|------|------------------|
| C8E2 | CA21 | C503 | C514 | 0C0D | 140F | 0700 | C8E2 | H.J!E.E.....H.   |
| CA21 | DF00 | 040A | 1CE0 | C536 | 8102 | 0700 | 8405 | J!.....E6.....   |
| 7C00 | 01D8 | 007A | 007C | 007B | 007D | 01D8 | 845C | ...X.....X.\     |
| 0CFB | 0700 | C8E2 | CA21 | C503 | C514 | 0CF3 | 1454 | ....H.J!E.E....T |

Step 10. Modify the sector with the patch. The modify command

and a display of the sector after modification are shown

below:

//MD 1 >6E66 >D >2102 >8102 >9405

//DD 1 >6E66  
6E66

|      |      |      |      |      |      |      |      |                  |
|------|------|------|------|------|------|------|------|------------------|
| C8E2 | CA21 | C503 | C514 | 0C0D | 140F | 0700 | C8E2 | H.J!E.E.....H.   |
| CA21 | DF00 | 040A | 1CE0 | C536 | 2102 | 8102 | 9405 | J!.....E6!.....  |
| 7C00 | 01D8 | 007A | 007C | 007B | 007D | 01D8 | 845C | ...X.....X.\     |
| 0CFB | 0700 | C8E2 | CA21 | C503 | C514 | 0CF3 | 1454 | ....H.J!E.E....T |

Step 11. The patch is now on the disc. Repeat steps 2 through

10 for all the patches.



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Table 1.  
Standard Operating System Patch Guide  
for the Diablo

System Revision Letter: \*B

Date: 18 Aug. 1975

| Patch # | Description                                                 | Module Name | Absolute Sector | Word Offset into Sector | Present        | Patch          |
|---------|-------------------------------------------------------------|-------------|-----------------|-------------------------|----------------|----------------|
| 1       | Multitasking special case causes crash code >F              | IOHEAD      | >6D3            | 3<br>4                  | >0F01<br>>8907 | >5400<br>>02E4 |
| 2       | User start job problem - go to READY but will not go to RUN | JUSTAR      | >D44            | 13                      | 0F01<br>890A   | 0F1A<br>8901   |



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Figure 1. Printout of Patches Applied to the Standard System

on a Diablo Disc

System Revision Letter: \*B

Date: 18 Aug. 1975

Patch 1

//DD 1 >6D3

06D3

|      |      |      |      |      |      |      |      |                  |
|------|------|------|------|------|------|------|------|------------------|
| 0F02 | C0E1 | 8908 | 0F01 | 8907 | 1F07 | C0E3 | 740A | ...Q.....Q...    |
| 0502 | C565 | 1900 | C536 | 8105 | C556 | 788D | 1F72 | ..E...E6..EV.... |
| 07D3 | 0F68 | 15A9 | B103 | C516 | C507 | 03E7 | 001F | .S...>1.E.E..... |
| 0002 | 000C | 0001 | 074A | 0000 | 0000 | 0000 | 0000 | .....J.....      |

//MD 1 >6D3 3 >5400 >02E4

//DD 1 >6D3

06D3

|      |      |      |      |      |      |      |      |                   |
|------|------|------|------|------|------|------|------|-------------------|
| 0F02 | C0E1 | 8908 | 5400 | 02E4 | 1F07 | C0E3 | 740A | ...Q...T.....Q... |
| 0502 | C565 | 1900 | C536 | 8105 | C556 | 788D | 1F72 | ..E...E6..EV....  |
| 07D3 | 0F68 | 15A9 | B103 | C516 | C507 | 03E7 | 001F | .S...>1.E.E.....  |
| 0002 | 000C | 0001 | 074A | 0000 | 0000 | 0000 | 0000 | .....J.....       |

Patch 2

//DD 1 >D44

0D44

|      |      |      |      |      |      |      |      |                  |
|------|------|------|------|------|------|------|------|------------------|
| 8103 | 0700 | C516 | 8115 | C556 | 0103 | C844 | C8C4 | ....E...EV..HDHD |
| 8102 | 0F01 | 8C0C | C0D1 | 890B | 0F01 | 890A | 1F0A | .....Q.....      |
| C0D3 | 7408 | 0F00 | 8C03 | B106 | C516 | C507 | 02E4 | QS.....1.E.E...  |
| 1F72 | 0074 | 1C96 | 000F | 0000 | 0000 | 0000 | 0000 | .....            |

//MD 1 >D44 13 >0F1A >8901

//DD 1 >D44

0D44

|      |      |      |      |      |      |      |      |                  |
|------|------|------|------|------|------|------|------|------------------|
| 8103 | 0700 | C516 | 8115 | C556 | 0103 | C844 | C8C4 | ....E...EV..HDHD |
| 8102 | 0F01 | 8C0C | C0D1 | 890B | 0F1A | 8901 | 1F0A | .....Q.....      |
| C0D3 | 7408 | 0F00 | 8C03 | B106 | C516 | C507 | 02E4 | QS.....1.E.E...  |
| 1F72 | 0074 | 1C96 | 000F | 0000 | 0000 | 0000 | 0000 | .....            |



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Table II  
Nonstandard Operating System Patch Guide

System Revision Letter: \*B

Date: 18 Aug. 1975

| Patch # | Load Module | Load Mod Start Sector LS | MIP # | Directory Sector Offset DS | Words Word Offset DW | Sector Offset Into MIP MS | Calc Abs Sector AS | Word Offset into Sector | Present        | Patch          |
|---------|-------------|--------------------------|-------|----------------------------|----------------------|---------------------------|--------------------|-------------------------|----------------|----------------|
| 1       | SYSLD       |                          | 0     |                            |                      | >4B                       |                    | 3<br>4                  | >0F01<br>>8907 | >5400<br>>02E4 |
| 2       | SYSLD       | >688                     | >6C   | >A                         | 2                    | 3                         |                    | 13<br>14                | >0F01<br>>890A | >0F1A<br>>8901 |

AS = LS + ( {LS + DS} sector, DW word ) + MS

AS = Absolute sector address of patch

LS = Absolute sector address of start of load module

DS = Directory words sector offset

DW = Directory words word offset

MS = Sector offset into MIP

( ) Signifies contents of particular word of particular sector (sector, word)



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Table 1.

Standard Operating System Patch Guide  
for the Diablo Disc

System Revision Letter: \*B  
Date: 29 Aug. 1975

| Patch # | Description                                                                                                                                                                                                                                  | Module Name | Absolute Sector | Word Offset into Sector | Present                                   | Patch                                     |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|-------------------------|-------------------------------------------|-------------------------------------------|
| 3       | Internal DMAC expansion system will not IPL with 2 <sup>d</sup> disc in system.<br>Note: 2 <sup>d</sup> disc may not be added to Device Table until this patch is made. After patch made to system, re IPL and then add 2 <sup>d</sup> disc. | SYSTBL      | >69E            | >4                      | >0000                                     | >4000                                     |
| 4       | DSCA OVERLOAD<br>CONDITION -JOB SUBMITTED<br>Message may not be printed or may cause subsystem to error off with I/O attempted on unassigned LUN.                                                                                            | JSCOMT      | >D28<br>>D28    | >7<br>>B                | >0F01<br>>F800<br>>B10E<br>>C516<br>>C507 | >0F02<br>>0808<br>>8914<br>>F800<br>>782F |



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Figure 1. Printout of Patches Applied to the Standard System

on a Diablo Disc

System Revision Letter: \*g

Date: 29 Aug. 1975

Patch 3

//DD 1 >69E

069E

|      |      |      |      |      |      |      |      |       |
|------|------|------|------|------|------|------|------|-------|
| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | ..... |
| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | ..... |
| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | ..... |
| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | ..... |

//MD 1 >69E 4 >4000

//DD 1 >69E

069E

|      |      |      |      |      |      |      |      |             |
|------|------|------|------|------|------|------|------|-------------|
| 0000 | 0000 | 0000 | 0000 | 4000 | 0000 | 0000 | 0000 | .....9..... |
| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | .....       |
| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | .....       |
| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000 | .....       |

Patch 4

//DD 1 >D28

0D28

|      |      |      |      |      |      |      |      |                   |
|------|------|------|------|------|------|------|------|-------------------|
| 8109 | 0700 | 840D | 1706 | C0E2 | C521 | 8913 | 0F01 | .....9.E!....     |
| 8912 | 1F12 | C0E3 | F800 | B10E | C516 | C507 | 00A1 | ....9...1.E.E!..  |
| 02E4 | 1E1B | 00A2 | 00A3 | 026A | 00A4 | 00A5 | 2503 | .....".#...\$.%%. |
| 040F | 2712 | 8102 | 0502 | 210A | 2F01 | 8502 | 0102 | ..!.....!./.....  |

//MD 1 >D28 >7 >0F02

//MD 1 >D28 >B >0808 >8914 >F800 >782F

//DD 1 >D28

0D28

|      |      |      |      |      |      |      |      |                   |
|------|------|------|------|------|------|------|------|-------------------|
| 8109 | 0700 | 840D | 1706 | C0E2 | C521 | 8913 | 0F02 | .....9.E!....     |
| 8912 | 1F12 | C0E3 | 0808 | 8914 | F800 | 782F | 00A1 | ....9.....!.      |
| 02E4 | 1E1B | 00A2 | 00A3 | 026A | 00A4 | 00A5 | 2503 | .....".#...\$.%%. |
| 040F | 2712 | 8102 | 0502 | 210A | 2F01 | 8502 | 0102 | ..!.....!./.....  |



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Table II  
Nonstandard Operating System Patch Guide

System Revision Letter: \*B

Date: 29 Aug. 1975

| Patch # | Load Module | Load Mod Start Sector LS          | MIP # | Directory Sector Offset DS | Words Word Offset DW | Sector Offset Into MIP MS | Calc Abs Sector AS | Word Offset into Sector | Present                                   | Patch                                     |
|---------|-------------|-----------------------------------|-------|----------------------------|----------------------|---------------------------|--------------------|-------------------------|-------------------------------------------|-------------------------------------------|
| 3       | SYSLD       | >688                              | ROOT  | N/A                        | N/A                  | >16                       |                    | >4                      | >0000                                     | >4000                                     |
|         |             | (This is location >96 in Memory.) |       |                            |                      |                           |                    |                         |                                           |                                           |
| 4       | SYSLD       | >688                              | >68   | 9                          | >16                  | 3                         |                    | >7<br>>B                | >0F01<br>>F800<br>>B10E<br>>C516<br>>C507 | >0F02<br>>0808<br>>8914<br>>F800<br>>782F |

AS = LS + ( {LS + DS} sector, DW word ) + MS

AS = Absolute sector address of patch

LS = Absolute sector address of start of load module

DS = Directory words sector offset

DW = Directory words word offset

MS = Sector offset into MIP

( ) Signifies contents of particular word of particular sector (sector, word)



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